

MKT 3434 – Machine Learning GUI Enhancement Report

**Safa Emre Beytekin
2206A020**

Introduction

This report presents the development and enhancement of a modular Machine Learning GUI, designed to support both classical and modern ML workflows. Originally built to train and evaluate models with basic data management functionality, the system has been significantly extended to incorporate dimensionality reduction techniques, evaluation metrics for clustering, cross-validation, and interactive visual analytics.

Summary of Initial Features

The original version of the GUI included the following features:

- Loading built-in datasets from scikit-learn (Iris, Breast Cancer, etc.)
- Manual CSV import and target column selection
- Classical ML models: Logistic Regression, SVM, KNN, Decision Trees, Random Forest
- Basic Deep Learning (MLP) architecture editor
- Matplotlib-based 2D result visualizations
- Data scaling and basic train/test split options

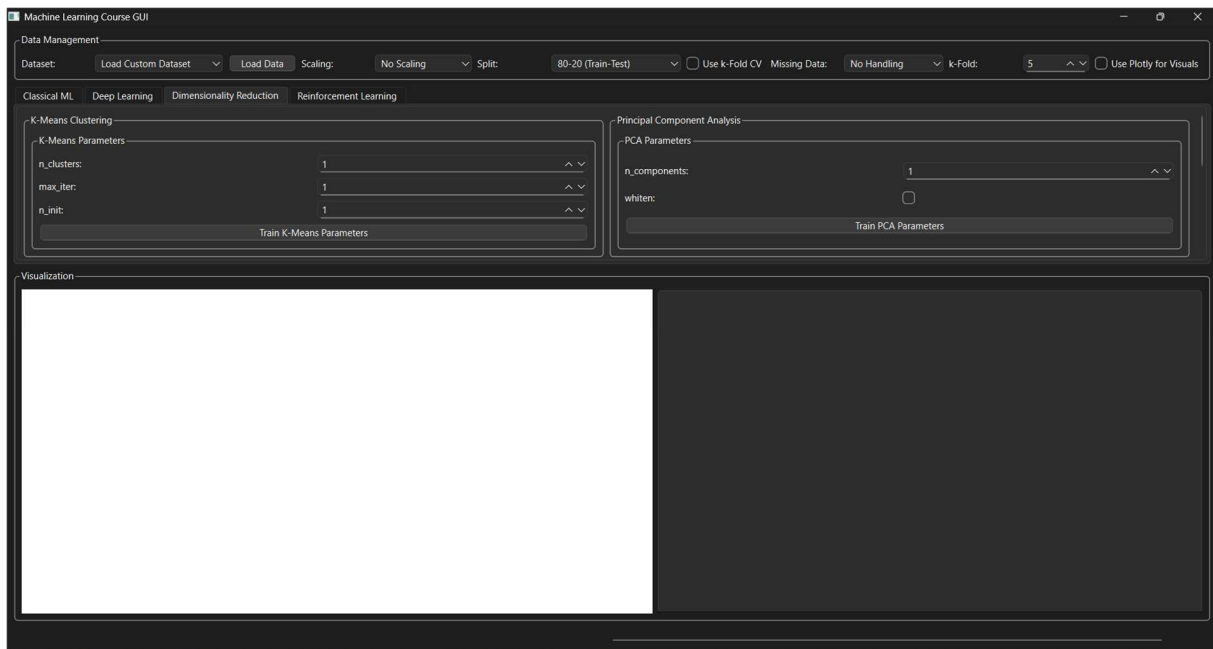


Figure 1: Original GUI layout with tabs and widgets.

Implemented Enhancements

a. Dimensionality Reduction Tab

A new tab was introduced to the GUI with support for four dimensionality reduction techniques:

- Principal Component Analysis (PCA)
- Linear Discriminant Analysis (LDA)

- t-distributed Stochastic Neighbor Embedding (t-SNE)
- Uniform Manifold Approximation and Projection (UMAP)

Each method includes parameter controls such as number of components, perplexity, and neighbors.

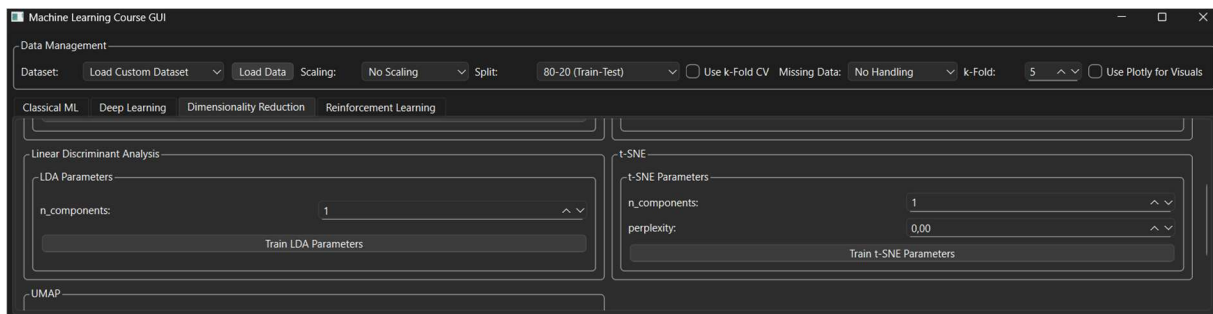


Figure 2: Dimensionality reduction tab and control panels.

b. Clustering Quality Metrics

- **KMeans Clustering:** Visual output of clusters overlaid on test data.
- **Evaluation Metrics:**
 - **Elbow Method** using inertia scores for various k values.
 - **Silhouette Score** to quantify clustering quality.

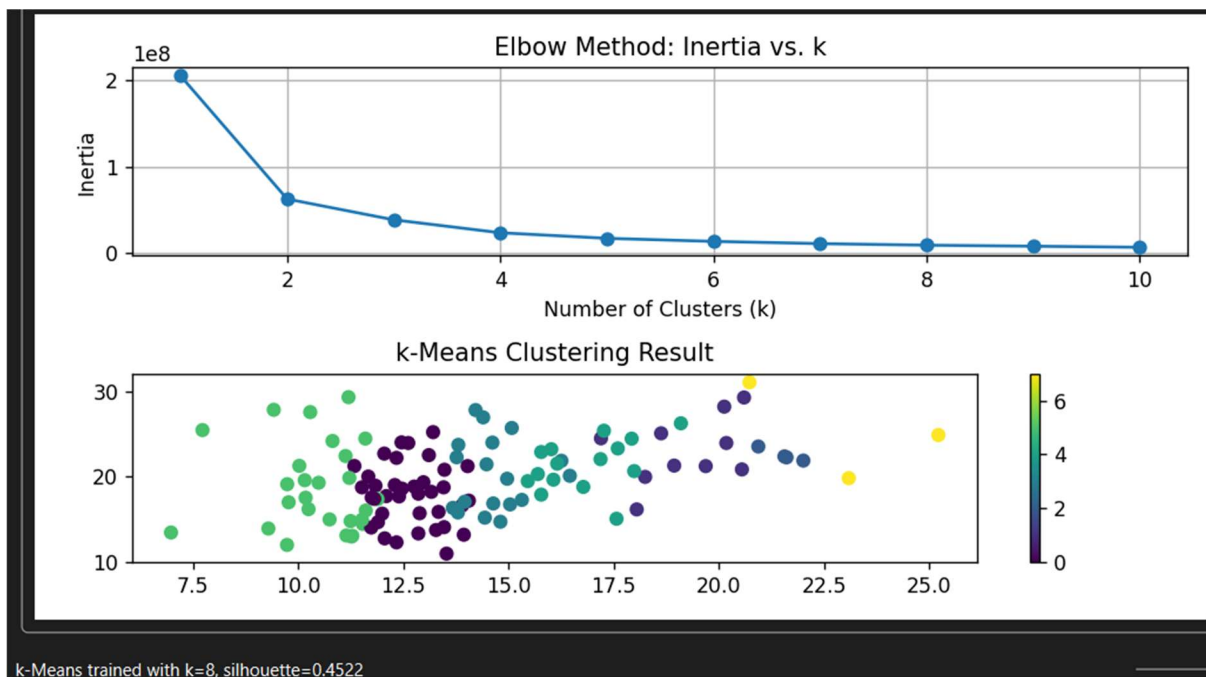


Figure 3: KMeans cluster visual and Elbow plot.

c. k-Fold Cross-Validation

Cross-validation capabilities were added via a simple toggle and user-defined k parameter (default: 5). Depending on model and loss function, different scorers are automatically selected:

- Accuracy for classification
- MSE/MAE for regression
- Cross-entropy and hinge loss where applicable

Resulting mean and standard deviation scores are shown to the user.

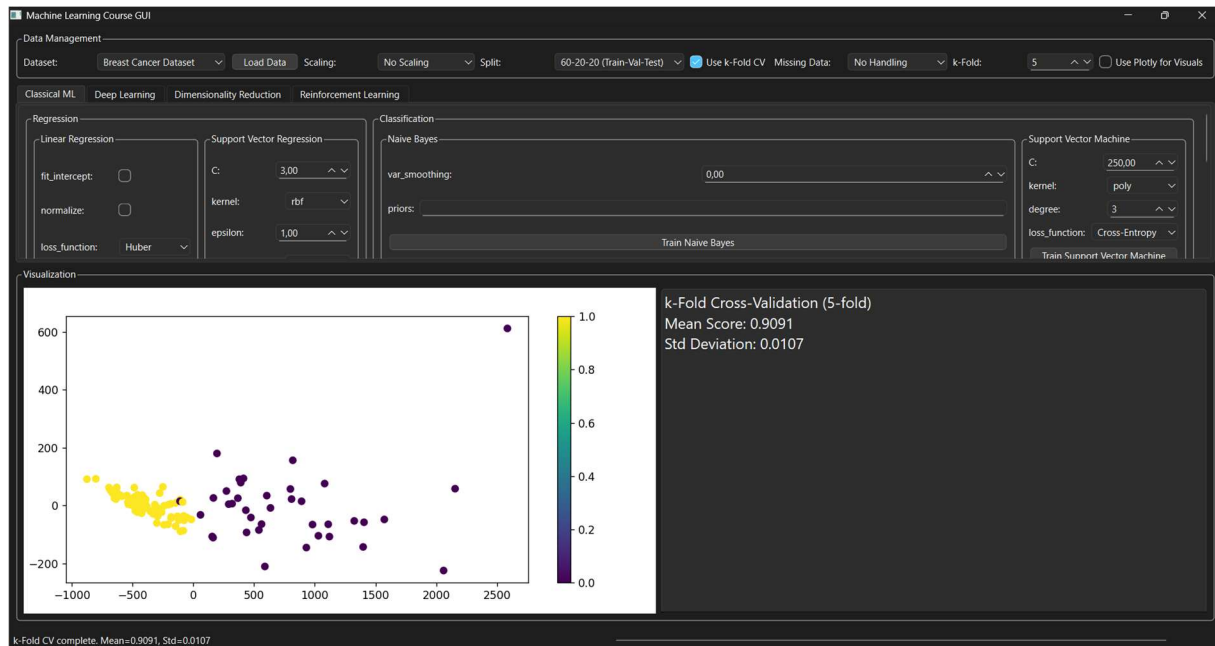


Figure 4: Panel showing cross-validation summary.

Dimensionality Reduction: Theory and Application

Method	Supervised	Main Purpose	Parameters	Visualization
PCA	No	Feature decorrelation and compression	n_components	2D/3D Scatter
LDA	Yes	Class separation and reduction	n_components	1D or 2D Axis
t-SNE	No	Non-linear embedding with local structure	perplexity, n_components	2D/3D
UMAP	No	Local and global structure preservation	n_neighbors, n_components	2D/3D

Each algorithm is implemented with appropriate GUI support and project-specific callbacks.

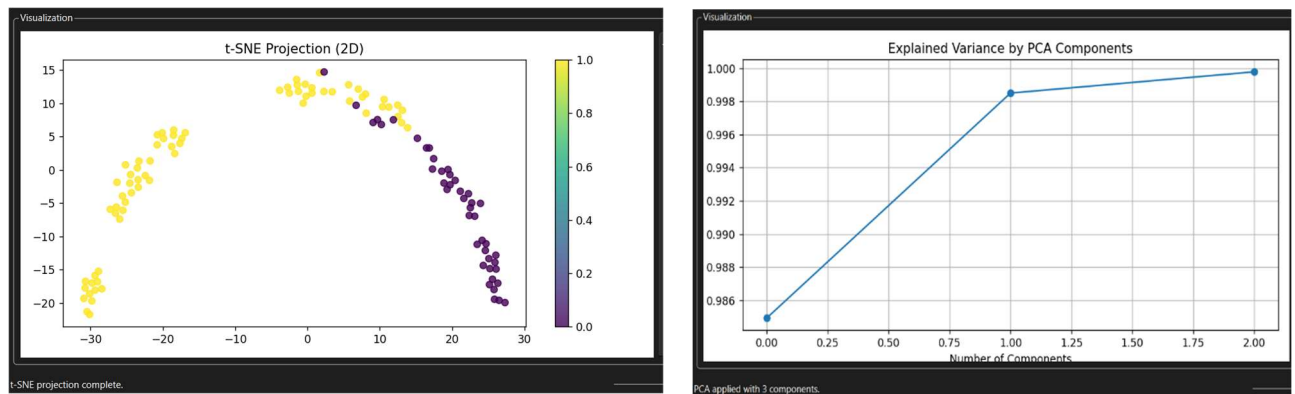


Figure 5: Side-by-side PCA and t-SNE results.

Clustering Evaluation Metrics

The GUI calculates **Silhouette Score** to assess how well-separated clusters are in reduced feature space.

This score is particularly relevant for KMeans and unsupervised dimensionality reduction outputs.

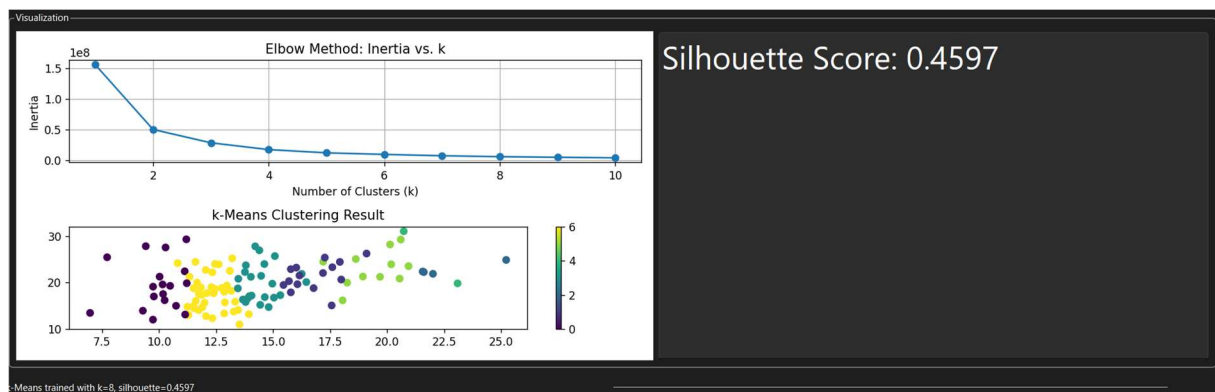


Figure 6: Silhouette score output after clustering.

Cross-Validation Support

Cross-validation is essential for reducing overfitting and validating generalization.

This system supports user-selected k-fold values between 2 and 20, with dynamic scorer selection.

Example:

- For k=5 and 100 samples, fold sizes will be [20, 20, 20, 20, 20]
- Output shows average accuracy or loss and standard deviation

Plotly Integration and Interactive Visualization

A checkbox enables Plotly mode for t-SNE and UMAP projections. When activated:

- Projections are displayed in plotly.express (2D or 3D automatically)
- Color-coded by class labels
- Responsive and zoomable charts for better visual exploration

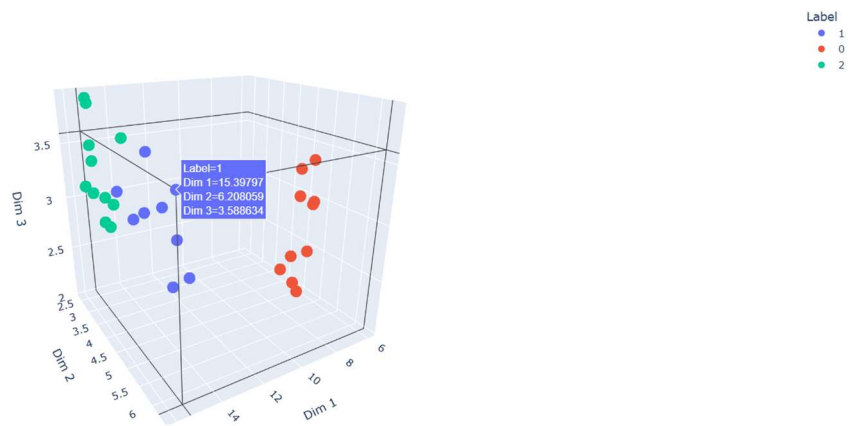


Figure 7: Plotly 3D scatter chart from UMAP.

8. Comparison of Dimensionality Reduction Methods

PCA offers fast linear projections but lacks non-linearity.

t-SNE is useful for discovering subtle clustering structure, though it is computationally heavier.

UMAP serves as a faster alternative to t-SNE with better preservation of global structures.

t-SNE Projection (3D)

UMAP Projection (3D)



Figure 8: Visual comparison of t-SNE vs UMAP on same dataset.

9. Conclusion

The Machine Learning GUI has been extended into a comprehensive educational tool that:

- Supports data preprocessing, model training, and evaluation
- Visualizes complex data transformations in 2D and 3D
- Allows experimentation with unsupervised clustering and feature compression
- Facilitates real-world applicable practices such as cross-validation and dimensionality reduction

Its modular structure ensures continued extensibility and usage in both research and teaching environments.